

Дифференциатор

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1 Результат дифференцирования

$$\left(\frac{\operatorname{arctg}(\cos(x)) - x}{\sin(x) - \operatorname{sh}(x)}\right)^{(x)^5} \cdot 5 \cdot (x)^4 \cdot \ln\left(\frac{\operatorname{arctg}(\cos(x)) - x}{\sin(x) - \operatorname{sh}(x)}\right) + (x)^5 \cdot \frac{\frac{-\sin(x)}{1+(\cos(x))^2} - 1 \cdot \sin(x) - \operatorname{sh}(x) - \operatorname{arctg}(\cos(x)) - x \cdot \cos(x) - \operatorname{ch}(x)}{(\sin(x) - \operatorname{sh}(x))^2}}$$

2 Пошаговое дифференцирование

$$\begin{aligned} 1. \quad & \frac{d}{dx}(\cos(x)) \\ &= -\sin(x) \end{aligned}$$

$$\begin{aligned} 2. \quad & \frac{d}{dx}(\operatorname{arctg}(\cos(x))) \\ &= \frac{-\sin(x)}{1+(\cos(x))^2} \end{aligned}$$

$$\begin{aligned} 3. \quad & \frac{d}{dx}(\operatorname{arctg}(\cos(x)) - x) \\ &= \frac{-\sin(x)}{1+(\cos(x))^2} - 1 \end{aligned}$$

$$\begin{aligned} 4. \quad & \frac{d}{dx}(\sin(x)) \\ &= \cos(x) \end{aligned}$$

$$\begin{aligned} 5. \quad & \frac{d}{dx}(\operatorname{sh}(x)) \\ &= \operatorname{ch}(x) \end{aligned}$$

$$\begin{aligned} 6. \quad & \frac{d}{dx}(\sin(x) - \operatorname{sh}(x)) \\ &= \cos(x) - \operatorname{ch}(x) \end{aligned}$$

$$\begin{aligned} 7. \quad & \frac{d}{dx}\left(\frac{\operatorname{arctg}(\cos(x)) - x}{\sin(x) - \operatorname{sh}(x)}\right) \\ &= \frac{\frac{-\sin(x)}{1+(\cos(x))^2} - 1 \cdot \sin(x) - \operatorname{sh}(x) - \operatorname{arctg}(\cos(x)) - x \cdot \cos(x) - \operatorname{ch}(x)}{(\sin(x) - \operatorname{sh}(x))^2} \end{aligned}$$

$$\begin{aligned} 8. \quad & \frac{d}{dx}((x)^5) \\ &= 5 \cdot (x)^4 \end{aligned}$$

$$\begin{aligned}
9. \quad & \frac{d}{dx} \left(\left(\frac{\operatorname{arctg}(\cos(x)) - x}{\sin(x) - \operatorname{sh}(x)} \right)^{(x)^5} \right) \\
&= \left(\frac{\operatorname{arctg}(\cos(x)) - x}{\sin(x) - \operatorname{sh}(x)} \right)^{(x)^5} \cdot 5 \cdot (x)^4 \cdot \ln \left(\frac{\operatorname{arctg}(\cos(x)) - x}{\sin(x) - \operatorname{sh}(x)} \right) + (x)^5 \cdot \frac{\frac{-\sin(x)}{1+(\cos(x))^2} - 1 \cdot \sin(x) - \operatorname{sh}(x) - \operatorname{arctg}(\cos(x))}{\frac{(\sin(x) - \operatorname{sh}(x))^2}{\operatorname{arctg}(\cos(x)) - x}}
\end{aligned}$$